

Lecture 4 Summary: Exploration–Exploitation Theory

Overview

The exploration–exploitation dilemma is central to reinforcement learning: how to balance gaining new information (exploration) versus maximizing known rewards (exploitation). This lecture formalizes that tradeoff through the *multi-armed bandit* (MAB) framework and explores provably optimal algorithms for regret minimization.

1. The Multi-Armed Bandit (MAB) Problem

A K -armed bandit has K independent actions (“arms”). At each time $t = 1, 2, \dots, T$:

- The learner selects an arm $a_t \in \{1, \dots, K\}$.
- Receives a random reward $r_t \sim \nu_{a_t}$ with mean μ_{a_t} .

Goal: maximize cumulative reward or equivalently minimize *regret*:

$$R_T = T\mu^* - \sum_{t=1}^T \mu_{a_t}, \quad \mu^* = \max_i \mu_i.$$

Regret quantifies the cost of exploration—how much reward is lost compared to always pulling the best arm.

2. Naïve Approaches

Greedy Algorithm. Always choose the arm with highest sample mean $\hat{\mu}_i$. Fails because it may prematurely exploit a suboptimal arm due to early noise.

ϵ -Greedy. With probability ϵ_t , explore randomly; otherwise exploit:

$$a_t = \begin{cases} \text{uniform random arm,} & \text{with prob. } \epsilon_t, \\ \arg \max_i \hat{\mu}_i, & \text{with prob. } 1 - \epsilon_t. \end{cases}$$

Typically ϵ_t decays over time (e.g., $\epsilon_t = 1/t$). Simple, but not optimal.

3. Optimism in the Face of Uncertainty

A more principled strategy: select arms using **Upper Confidence Bounds (UCB)**.

UCB1 Algorithm (Auer et al., 2002). At time t , choose

$$a_t = \arg \max_i \left(\hat{\mu}_i + \sqrt{\frac{2 \log t}{n_i}} \right),$$

where n_i is the number of times arm i has been pulled.

This embodies the idea of *optimism under uncertainty*: each arm's estimate is inflated by a confidence bonus that decreases as n_i grows.

Regret Bound.

$$\mathbb{E}[R_T] = O\left(\sum_{i:\Delta_i>0} \frac{\log T}{\Delta_i}\right), \quad \Delta_i = \mu^* - \mu_i.$$

This is provably near-optimal (Lai & Robbins, 1985).

4. Thompson Sampling (Bayesian Bandit)

A Bayesian alternative maintains a posterior $p(\mu_i|\text{data})$ for each arm:

- (a) Sample $\tilde{\mu}_i \sim p(\mu_i|\text{data})$ for all i .
- (b) Choose $a_t = \arg \max_i \tilde{\mu}_i$.
- (c) Update posterior with observed reward.

This balances exploration and exploitation automatically, achieving the same order of regret as UCB.

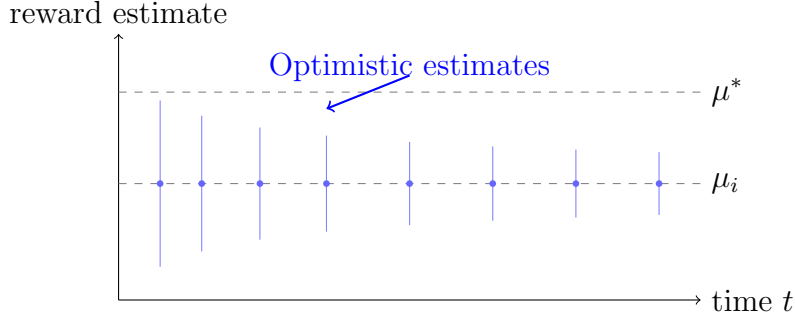


Figure 1: Illustration of UCB exploration: each arm’s reward estimate is augmented by a shrinking confidence interval.

5. Lower Bounds on Regret

Lai & Robbins Bound (1985). Any consistent algorithm satisfies:

$$\liminf_{T \rightarrow \infty} \frac{\mathbb{E}[R_T]}{\log T} \geq \sum_{i: \mu_i < \mu^*} \frac{\Delta_i}{D_{\text{KL}}(\nu_i \parallel \nu^*)}.$$

No algorithm can achieve asymptotically smaller regret.

6. Beyond the Classical Bandit

Contextual Bandits. At each step t , an observable context x_t is given, and rewards depend on both x_t and chosen action a_t . Policies $\pi(a|x)$ map contexts to actions.

Linear Bandits. Rewards follow $r_t = x_t^\top \theta^* + \eta_t$ with $\|\theta^*\|_2 \leq 1$. The OFUL (Optimism in the Face of Uncertainty for Linear models) algorithm chooses

$$a_t = \arg \max_a \left(x_{a,t}^\top \hat{\theta}_t + \alpha_t \|x_{a,t}\|_{V_t^{-1}} \right),$$

where $V_t = \sum_{\tau < t} x_{a_\tau, \tau} x_{a_\tau, \tau}^\top + \lambda I$. Regret scales as $O(d\sqrt{T \log T})$ for d -dimensional features.

Continuous vs. Discrete Action Spaces

The classical bandit algorithms (e.g., ϵ -greedy, UCB, Thompson Sampling) are defined for **discrete** action spaces—a finite number of arms that can each be sampled and estimated individually. In **continuous** action spaces, these approaches cannot directly apply since there are infinitely many possible actions.

To extend exploration–exploitation ideas to continuous spaces, additional *structure* on the reward function must be assumed.

1. Linear and Generalized Linear Bandits. If rewards satisfy a linear model

$$\mathbb{E}[r_t \mid a_t] = a_t^\top \theta^*,$$

then we can generalize UCB and Thompson Sampling by estimating θ^* . Algorithms such as **OFUL** (Optimism in the Face of Uncertainty for Linear models) or **Linear Thompson Sampling** select actions that balance predicted reward $a_t^\top \hat{\theta}_t$ with an uncertainty term $\|a_t\|_{V_t^{-1}}$.

2. Kernelized or Gaussian Process Bandits. When rewards vary smoothly with a , one can model

$$r(a) \sim \mathcal{GP}(m(a), k(a, a')),$$

using a Gaussian Process prior. The **GP-UCB** algorithm chooses

$$a_t = \arg \max_a \mu_{t-1}(a) + \beta_t^{1/2} \sigma_{t-1}(a),$$

where μ_{t-1} and σ_{t-1} are the posterior mean and uncertainty. This extends optimism-based exploration to continuous spaces.

3. Connection to Reinforcement Learning. In full RL with continuous actions, exploration occurs through **stochastic policies** $\pi_\theta(a|s)$ that inject noise directly into the action distribution. Algorithms such as policy gradient and actor–critic methods therefore embody a continuous analogue of bandit exploration, using entropy regularization or Gaussian noise to encourage diversity in actions.

Summary Table.

Setting	Algorithm Family	Exploration Mechanism	Continuous Actions?
Finite arms	UCB, Thompson, ϵ -greedy	Optimism or randomness	No
Linear rewards	OFUL, Linear TS	Confidence ellipsoids in $\mathbb{R}^d$	Yes
Smooth rewards	GP-UCB, GP-TS	Function-space uncertainty	Yes
Reinforcement Learning	Actor-Critic, Policy Gradient	Stochastic policy sampling	Yes

In summary, continuous-action extensions of bandit algorithms require exploiting structure—linearity, smoothness, or parametric policy forms—to generalize across infinitely many possible actions.

7. Key Takeaways

- Exploration is essential to learn accurate estimates; too little causes bias, too much causes inefficiency.
- UCB provides a frequentist confidence-based approach with logarithmic regret.
- Thompson sampling provides a Bayesian counterpart with similar guarantees.
- Contextual and linear bandits generalize exploration–exploitation to structured, high-dimensional settings.